MILITARY SPECIFICATION

CABLE ASSEMBLIES: 7W3 and T4182

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of cable assembly entitled 7W3 Adapter, Case and T4182 Adapter, Test for the XM91 Permissive Action Link (P.A.L.) which is used on the Sergeant warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification For.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedures For.

DRAWINGS

ORDNANCE CORPS

8832984 - Cable Assembly, T4182 Adapter, Test.

FSC: 1135

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P1-R	J1-R	0.1 max
P1-S	J1-S	0.1 max
P1-U	J1-U	0.1 max

3.6 Insulation resistance. The insulation resistance between each terminal and all other terminals listed for the cable assemblies in Table II shall be greater than 20 megohms when tested at 500 ± 50 vdc as specified in 4.4.2. Unused terminals shall be connected together and tested as one terminal.

Table II
7W3 Case Adaptor, T4182 Test Adaptor
Insulation Resistance Test

P1-Shell,A,B,C,D,E,F,G,H,J,K,L,M,P,R,S,U
P1-Unused terminals: T & N
J1-Unused terminals: T & N

3.7 Environmental testing. The cable assemblies shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The cable assemblies shall also comply with applicable paragraphs of 3 before and after each environmental test. Testing shall be as specified in 4.4.4. Tests shall be conducted in the order specified.

3.7.1 High temperature. The cable assemblies shall show no evidence of damage or deterioration such as distortion, cracks, or swelling when subjected to +160 ± 5 degrees Fahrenheit (°F) as specified in 4.4.4.1 and shall meet the requirements of 3.6 while at this temperature. The cable assemblies shall meet the requirements of 3.5 after completion of high temperature test and return to room ambient (+80°F ± 2°F).

3.7.2 Low temperature. The cable assemblies shall show no evidence of damage or deterioration such as distortion, cracks, or swelling when subjected to -65°F ± 5°F as specified in 4.4.4.2 and shall meet the requirements of 3.6 while at this temperature. The cable assemblies shall meet the requirements of 3.5 after completion of low temperature test and return to room ambient (+80°F ± 20°F).

3.7.3 Humidity. The cable assemblies shall withstand humidity of 95 ± 5 percent and maximum temperature of +160 ± 5°F without evidence of damage or deterioration such as distortion, cracks or swelling when tested as specified in 4.4.4.3. The cable assemblies shall meet the requirements of 3.6 within one hour after removal from humidity chamber. The cable assemblies shall meet the requirements of 3.5 after completion of low temperature test and return to room ambient (+80°F ± 20°F).

3.7.4 Transportation vibration.

3.7.4.1 T4182 adaptor, test. The assembly shall withstand vibration for 45 ± 1 minutes in the vertical plane only (vertical plane being defined as that plane in which

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cold shuts, inclusions, porosity, or any similar characteristic.

3.9.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.9.3 Burrs. Parts shall be free from burrs.

3.9.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.9.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.9.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 3 complete assemblies (see dwg 8832984 and 8881470) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table IV . Tests shall be conducted in order specified in Table IV.

Table IV

First article inspection and tests

Inspection or Test	Dest	Non-Dest	Testing Sequence 1	2	3
1. Classification of characteristics (see 4.3.2.1)		X	X	X	X

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Table V
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Major:		
101.	Lead wire not completely covered by molded silicone	Visual
102.	General damage	Visual
103.	Incorrect length	Scale
*104.	Improper height of "male" terminals (pins of the same diameter must all appear to be within 1/32 inch in height relative to the adjacent pins)	Visual/Gage
**105.	Improper depth of "female" socket contact spring (female sockets of the same diameter must all appear to be within 1/32 inch in depth relative to the adjacent sockets)	Visual/Gage
106.	Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
201.	Incorrect or illegible cable and connector markings	Visual
202.	Poor bond between molded silicone and tubing and/or connector shell	Visual
203.	Voids and air pockets greater than 1/8" diameter in silicone	Visual

*In the event a pin does not appear to be within 1/32 inch in height relative to the adjacent pin, a gage (see 4.3.4) shall be used to determine the variation from the normal height. Reject any pin that varies by more than 0.032 inch maximum from the normal pin height of the individual connector. The normal height of the individual connector pin shall be determined by calculating the average height from measurements taken of pins adjacent to a pin that appears defective. The forward metal edge of the connector shall be utilized as the reference surface for all measurements.

**In the event a socket insert does not appear to be within 1/32 inch in depth relative to an adjacent socket, a gage (see 4.3.3) shall be used to determine the variation from the normal depth. Reject any insert that varies by more than 0.032 inch maximum from the normal socket insert depth of an individual connector. The normal depth of an individual connector socket shall be determined by calculating the average depth from measurements taken of sockets adjacent to the socket that appears defective. The forward metal edge of the connector shall be utilized as the reference surface for all measurements.

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4.3.3.3 High temperature, low temperature and humidity. The first cable made from each new lot of cable outer sheathing and/or molding compound shall be tested in accordance with 4.4.4.1, 4.4.4.2 and 4.4.4.3. If any cable fails to comply with the 3.7.1, 3.7.2 and 3.7.3, the cable as well as the entire lot of tubing or molding compound shall be rejected.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL). Els for assemblies are listed on the Equipment Tabulation, ET-8832984 and 8881470. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Mating assembly . The cable connectors will be mated to a fixture equivalent to the terminations they are to be used on.

4.4.2 Insulation resistance. The insulation resistance between each terminal shall be measured with a megohmmeter, or Ordnance approved test circuit device, at 500 + 50 volts direct current (vdc). The insulation resistance shall not be less than 20 megohms as specified in 3.6. Any circuit exhibiting an insulation resistance of less than 20 megohms shall not be rejected until the reading has been verified by the application of the specified voltage for 60 + 5 seconds. If, at the end of this time period, the insulation resistance has increased to at least 20 megohms, the circuit shall be accepted.

4.4.3 Circuit resistance. The resistance between the pins listed in 3.5 shall be measured with a Government approved electrical resistance tester. The resistance shall not exceed the value specified in 3.5.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The cable assemblies shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Level C. The cable assemblies shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.2 Packing.

5.2.1 Levels A and C. The cable assemblies shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Supersession data. This specification supersedes the requirements of Purchase description PA-PD-2498 dated 14 December 1967.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1135-A181

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